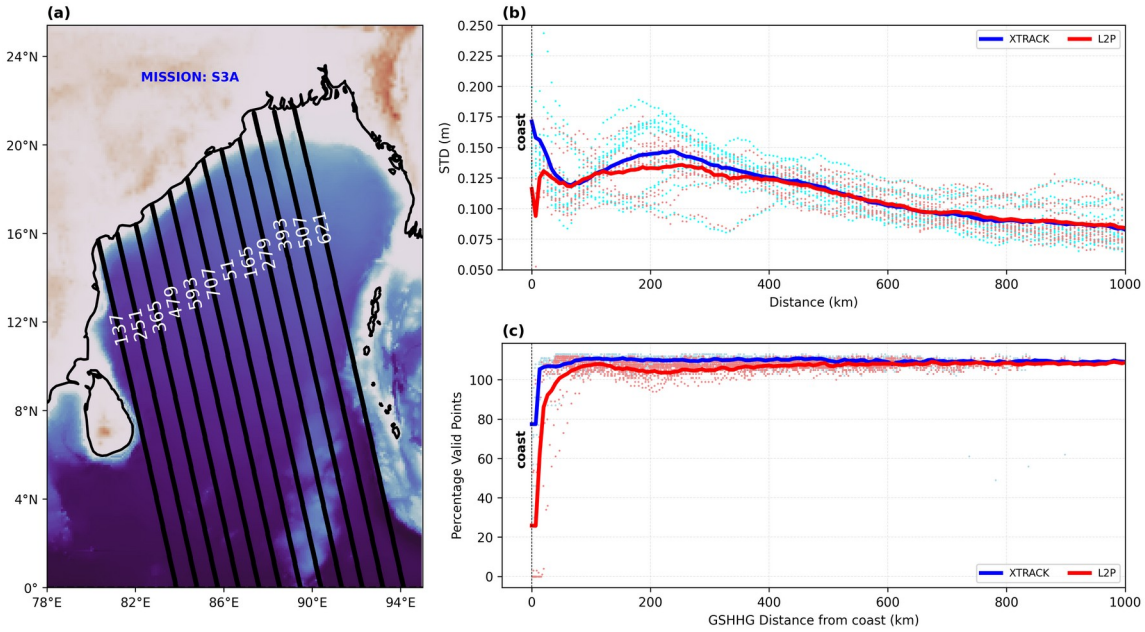


Fig1: Comparison of Sentinel-3A (S3A) sea level anomaly (SLA) data between X-TRACK and along-track Level-2+ (L2P) products for common ground tracks that are meeting the east coast of India. (a) Selected common S3A tracks plotted (black) over ETOPO5 bathymetry, with track numbers annotated in white. (b) Temporal standard deviation of SLA as a function of distance from coast for individual tracks (dots) and their multi-track mean (solid lines) from X-TRACK (blue) and L2P (red) products. (c) Percentage of valid SLA observations available along-track plotted against GSHHG coastline distance, showing X-TRACK (blue) and L2P (red) individual track values (dots) and multi-track means (solid lines). The dashed vertical line in (b) and (c) marks the coastline.

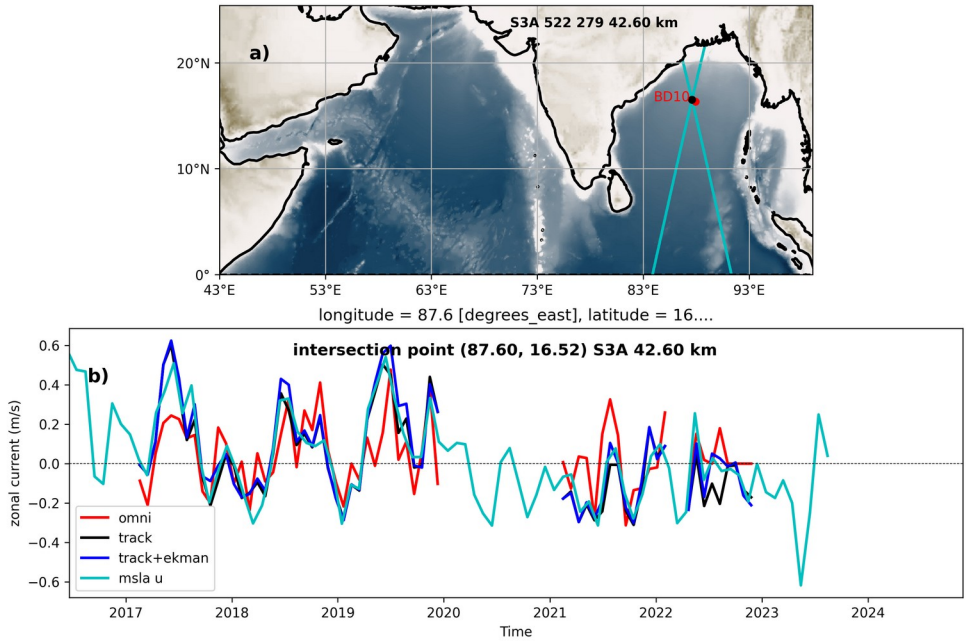


Fig2: b) Time series of geostrophic zonal component (black) derived at an intersection point (of track numbers 522 and 279 of S3A satellite mission) which is close to an OMNI buoy (BD10) overlied with buoy's zonal current(red). Timeseries of gu correted with ekman u component(blue) and AVISO msla at the intersection point(cyan) also added. a) Shows the intersection point(black) of the tracks and the close by bouy(red). It is rougly half a degree away from the intersection point.

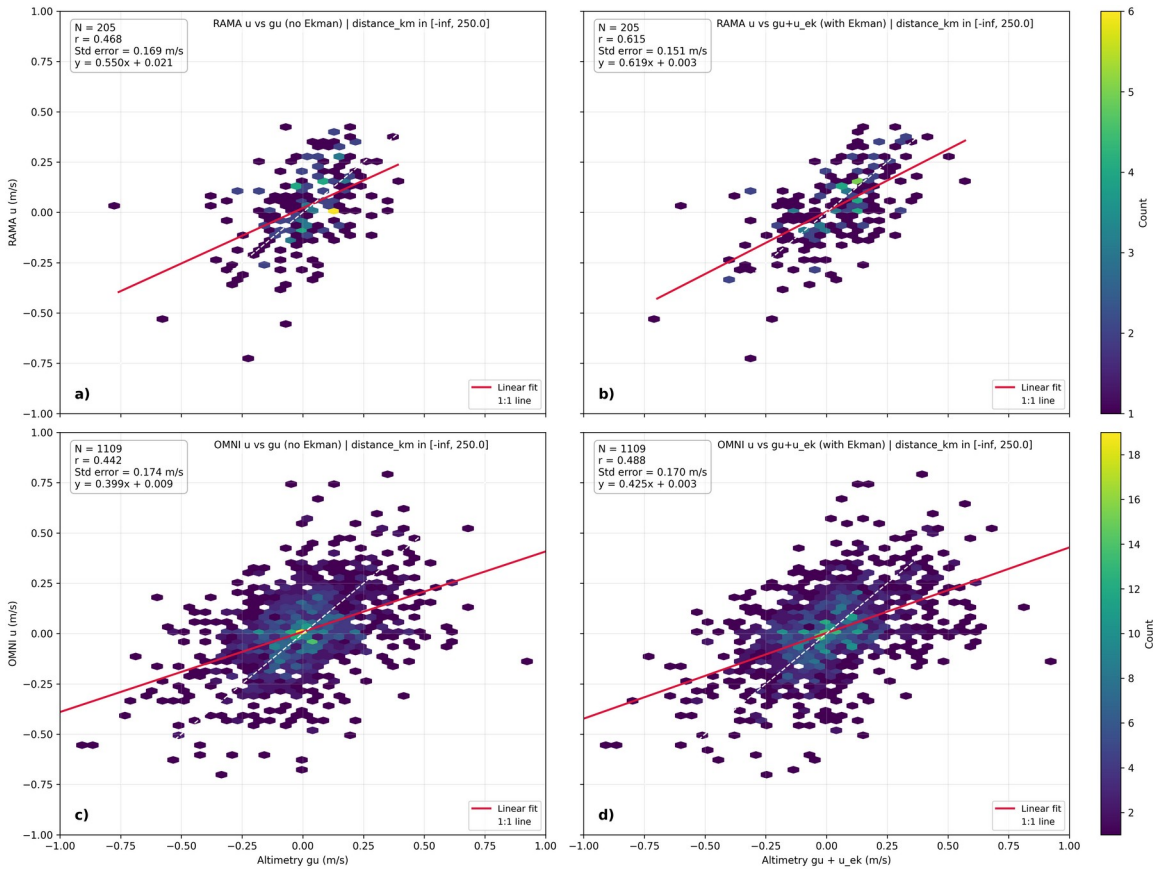


Fig3: a) Scatter plot of rama buoys' zonal currents(y-axis) versus altimetry derived zonal currents(x-axis) at nearby intersection points. b) Shows scatter plot with altimetry derived currents corrected with ekman zonal currents. c) and d) show scatter plots of OMNI currents with and without Ekman correction against altimetry zonal currents. The colorbar shows density of points in the scatter plot in terms of count. A redline passing through the points indicates linear fit and white dashed line represents perfect fit as a reference. The number of points, correlations and standard errors are show in boxes in the upper left corner.

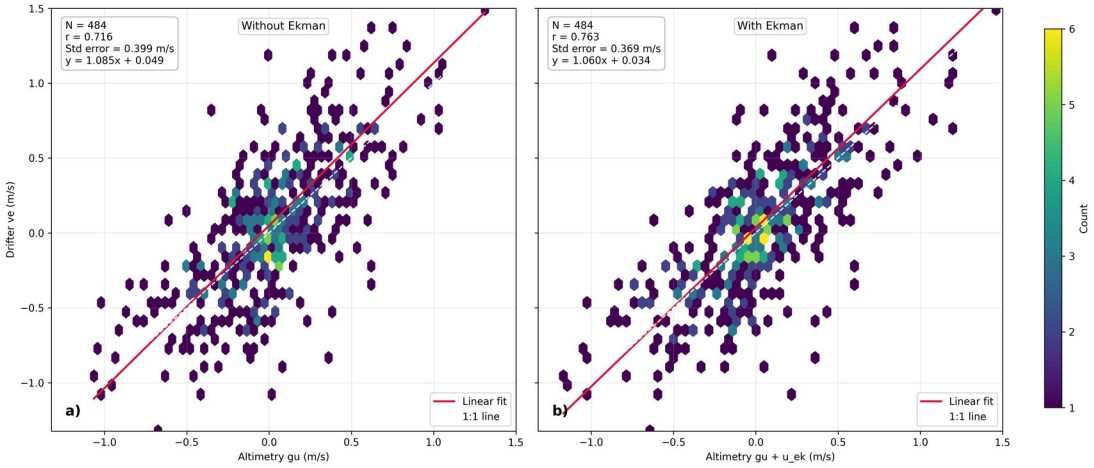


Fig4: a) Scatter plot of drifters' zonal velocities versus altimetry derived velocities within 200km inshore of the coast for the satellite mission TP+J1+J2+J3+S6A. b) shows scatter plot with altimetry derived velocities corrected with ekman zonal velocities.The colorbar shows density of points in the scatter plot in terms of count. A redline passing through the points indicates linear fit and white dashed line represents perfect fit as a reference. The number of points, correlations and standard errors are show in boxes in the upper left corner.

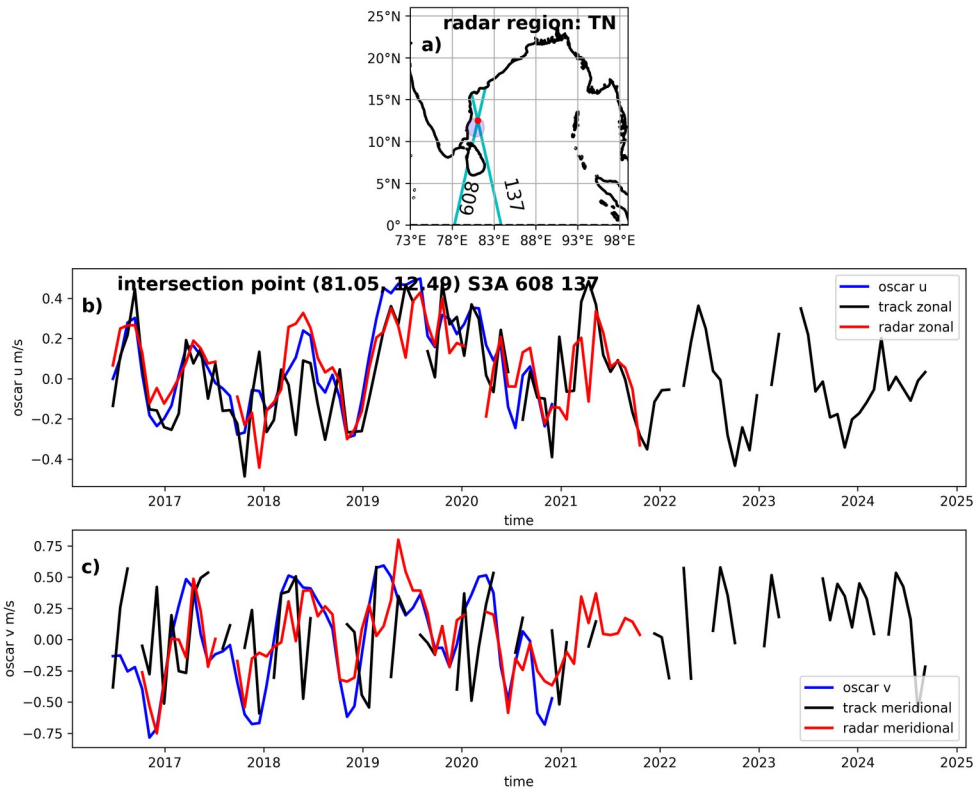


Fig5: a) Shows the radar region of TN with a intersection point within the region. b) and c) show zonal and meridional currents of altimetry derived geostrophic currents at the intersection point along with radar zonal velocity time series.

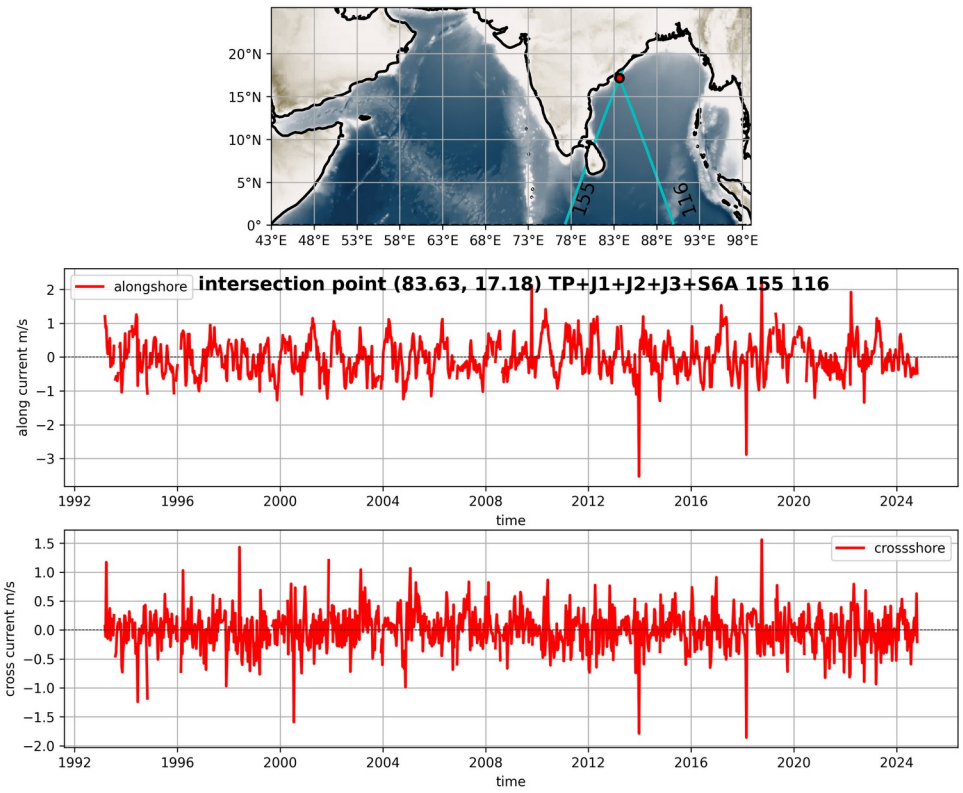


Fig6: a) An intersection point very close to the east coast of India belonging to the satellite mission TP+J1+J2+J3+S6A. b) and c) show thirty year timeseries of along shore and across shore current components at the intersection of two satellite tracks (155, 116).

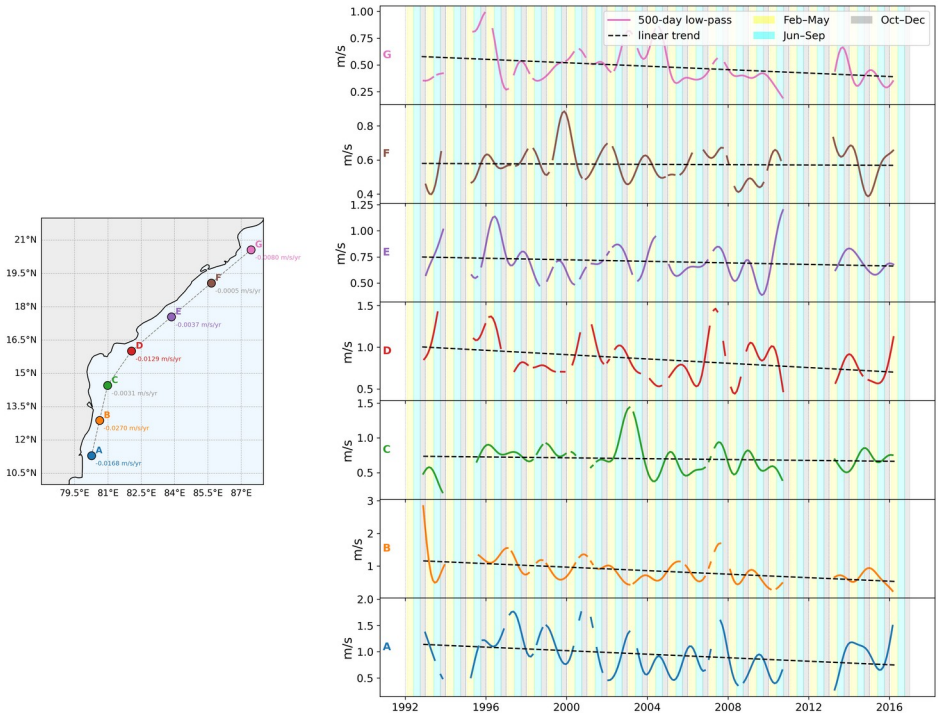


Fig7: Fig. X. Long-term variability and linear trends of geostrophic current speed at seven crossover points of the East India Coastal Current (EICC) derived from ERS-1/2, Envisat, and SARAL/AltiKa altimetry. (Left) Locations of the seven crossover points (A–G, south to north) along the eastern boundary of the Bay of Bengal (10°–22°N, 78°–88°E). Each point is colour-coded by its label; the annotated value is the linear trend in m s^{-1} yr^{-1} (greyed out where the trend is not significant at the 95% confidence level, $p \geq 0.05$). (Right) 500-day low-pass filtered current speed (coloured line) and its linear trend fit (black dashed line) at each point. Background shading indicates the spring inter-monsoon (February–May, yellow), summer monsoon (June–September, cyan), and post-monsoon/early winter (October–December, grey) seasons. Point coordinates and contributing ground-track pairs are: A (80.26°E, 11.30°N; T868×T567), B (80.62°E, 12.89°N; T868×T23), C (80.98°E, 14.46°N; T868×T481), D (82.06°E, 16.01°N; T324×T395), E (83.86°E, 17.55°N; T238×T767), F (85.65°E, 19.07°N; T152×T137), and G (87.45°E, 20.57°N; T66×T509).

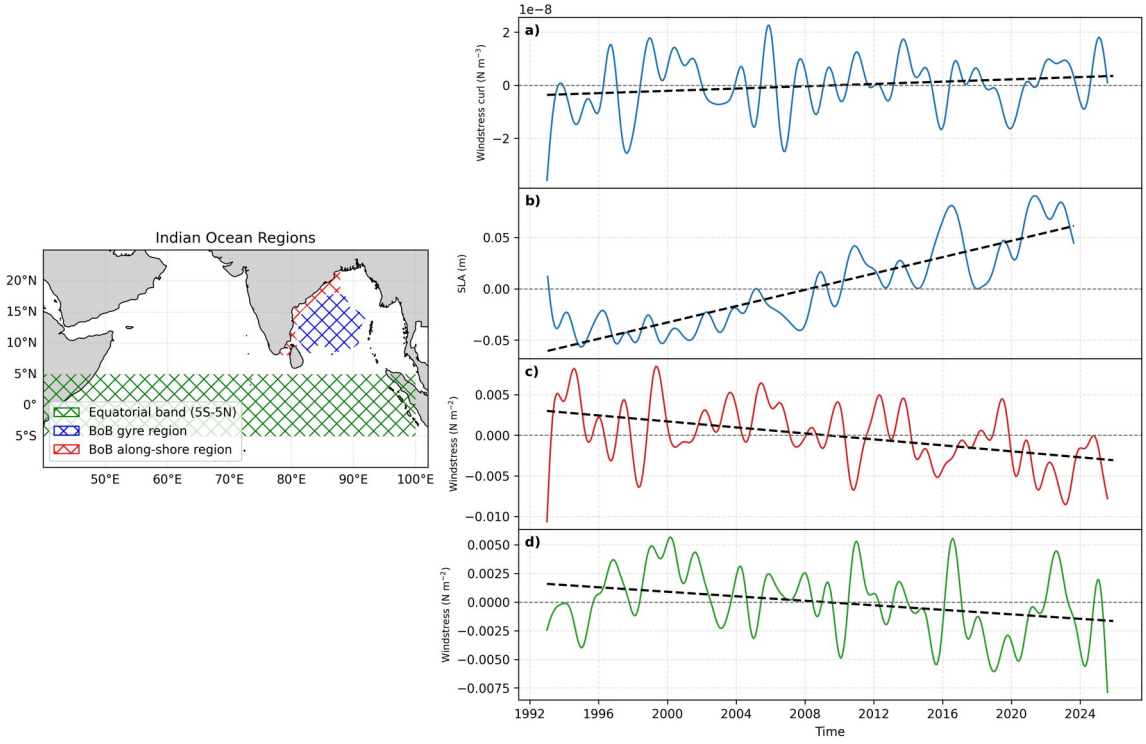
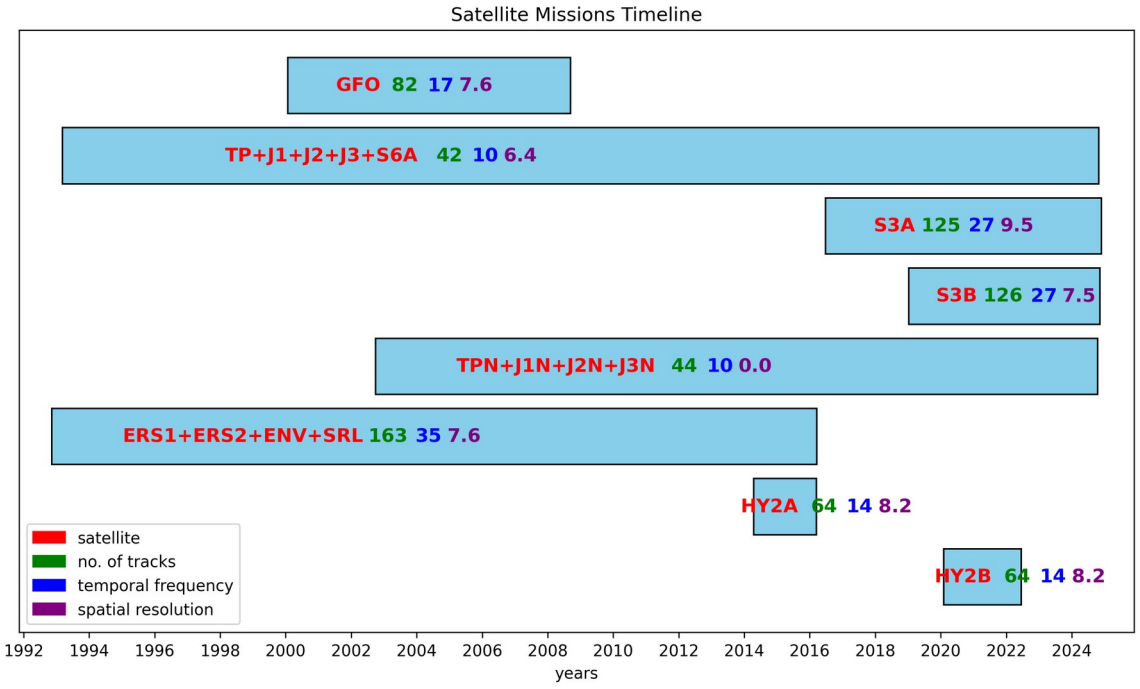

Indian Ocean Regions
20°N
15°N
10°N
5°N
0°
5°S
50°E
60°E
70°E
80°E
90°E
100°E
Equatorial band (5S-5N)
BoB gyre region
BoB along-shore region
1e−8
a)
Windstress curl (N m⁻³)
b)
SLA (m)
c)
Windstress (N m⁻²)
d)
Windstress (N m⁻²)
1992
1996
2000
2004
2008
2012
2016
2020
2024
Time

Fig8



Sup Fig1: Total number of coastal altimetry tracks for the region NINDIAN, temporal and spatial resolution of reference tracks for all the satellite missions.